Precision Care Challenge 2026

AI-Driven Prioritization System for Early Alzheimer's Diagnostic Pathways

Problem Statement

Alzheimer's disease is a progressive neurodegenerative condition where early detection plays a critical role in enabling timely intervention and improving patient outcomes. However, in real-world clinical settings, large populations of patients undergo initial cognitive screening (such as MoCA or similar assessments), and only a small subset progresses through further diagnostic pathways due to constraints in time, cost, and availability of advanced testing such as MRI and PET scans.

Currently, the process of deciding which patients should undergo additional diagnostic tests relies heavily on manual assessment and clinical judgment. This can lead to delays in identifying high-risk individuals, inefficient use of medical resources, and missed opportunities for early intervention.

There is a need for a system that can leverage existing patient data—including cognitive assessments, co-morbidities, blood work, and imaging—to identify patients who are at higher risk of developing Alzheimer's disease and should be prioritized for further diagnostic evaluation.

Such a system would enable a structured, step-wise diagnostic workflow, ensuring that high-risk patients are identified early and advanced diagnostic resources are utilized efficiently.

Challenge Objective

The objective of this challenge is to design and build a decision-support system that analyzes existing patient data to prioritize individuals for early and progressive Alzheimer's diagnostic evaluation.

The system should:

- Identify high-risk patients from large screened populations
- Recommend a structured escalation pathway for further diagnostic testing
- Support neurologists in accelerating diagnosis for patients most likely to benefit from early intervention

The solution must function strictly as a clinical decision-support tool and must not provide medical diagnoses or treatment recommendations.

Expected Solution Capabilities

1. Data Ingestion and Integration

- Use publicly available datasets such as:
  - ADNI cohort
  - OASIS cohort
- Ingest patient-level data including:
  - Cognitive test scores (e.g., MoCA or MMSE)
  - Co-morbidities
  - Blood test results
  - MRI data
- Organize data into a unified patient-centric structure

2. Risk-Based Patient Prioritization

- Analyze patient data to generate a risk-based prioritization score
- Identify individuals with a higher probability of progressing toward Alzheimer's disease
- Enable ranking or stratification **(e.g., high, medium, low priority)** with reasoning

3. Progressive Diagnostic Pathway Recommendation

Implement a multi-stage clinical escalation framework:

Stage 1: Cognitive Screening

- Input: Cognitive tests + baseline clinical data
- Output: Initial risk prioritization

Stage 2: Blood-Based Biomarkers

- Recommend follow-up testing for higher-risk patients
- Refine prioritization

Stage 3: MRI Evaluation

- Further assess structural brain changes

- Narrow down high-risk candidates

Stage 4: PET Scan Prioritization

- Identify patients suitable for advanced diagnostic confirmation
- Highlight potential treatment-eligible individuals

4. Clinician-Focused Interface

- Provide a simple interface for neurologists
- Display:
  - Prioritized list of patients
  - Suggested next diagnostic steps
  - Patient progression through diagnostic stages
- Ensure outputs are clear and easy to interpret

5. Explainability and Transparency

- Provide reasoning for patient prioritization
- Highlight contributing factors (e.g., cognitive decline, clinical indicators)
- Ensure the system remains interpretable and clinically understandable

6. Security and Ethical Considerations

- Use only public, open, or synthetic datasets
- Do not use identifiable patient information
- Clearly communicate system limitations
- Ensure the system acts only as a support tool and not a diagnostic authority

Data Guidance

Teams must use openly available datasets such as ADNI and OASIS or other public sources. No proprietary or confidential patient data may be used.

Reference Cloud Architecture (High-Level)

- Patient data from cognitive screening and clinical records is ingested
- Data is standardized and linked at the patient level
- A risk-scoring model identifies high-risk individuals
- Patients are prioritized into a structured diagnostic pipeline:
  - Cognitive → Blood Biomarkers → MRI → PET
- A clinician-facing interface presents prioritized patients and recommended next steps

Evaluation Criteria

- ***Clinical Relevance (20%)***
  - Ability to meaningfully prioritize patients for early diagnosis
- ***Data Handling (20%)***
  - Effective integration and use of multimodal patient data
- ***Prioritization Logic (20%)***
  - Clarity, interpretability, and validity of risk scoring
- ***User Experience (15%)***
  - Simplicity and usability for neurologists
- ***Ethics and Safety (10%)***
  - Responsible, non-diagnostic usage and transparency
- ***Innovation and Impact (15%)***
  - Potential to accelerate diagnosis and optimize healthcare resources.

Outcome

This challenge aims to explore how AI-driven decision-support systems can transform early Alzheimer's detection by converting large-scale screening data into actionable clinical prioritization.

The expected outcome is a system that helps neurologists:

- Identify high-risk patients earlier
- Optimize use of advanced diagnostic tools

- Accelerate the pathway to diagnosis and potential treatment

Ultimately, the solution should demonstrate how existing data can be leveraged to improve both patient outcomes and healthcare efficiency in neurodegenerative disease management.